# Unsupervised Domain Adaptation With Hierarchical Masked Dual-Adversarial Network for End-to-End Classification of Multisource Remote Sensing Data

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***Abstract*— Although unsupervised domain adaptation (UDA) has been successfully applied for cross-scene classification of multisource remote sensing (MSRS) data, there are still some tough issues: 1) the vast majority of them are patch-based, requiring pixel-by-pixel processing at high complexity and ignoring the roles of unlabeled data between different domains and 2) traditional masked autoencoder (MAE)-based methods lack effective multiscale analysis and require pre-training, ignoring the roles of low-level representations. As such, a hierarchical masked dual-adversarial DA network (HMDA-DANet) is proposed for cross-domain end-to-end classification of MSRS data. First, a hierarchical asymmetric MAE (HAMAE) without pre-training is designed, containing a frequency dynamic large-scale convolutional (FDLConv) block to enhance important structural information in the frequency domain, and an intramodality enhancement and intermodality interaction (IAEIEI) block to embed some additional information beyond the domain distribution by expanding the cross-modal reconstruction space. Representative multimodal multiscale features can be extracted, while to some extent improving their generalization to the target domain (TD). Then, a multimodal multiscale feature fusion (MMFF) block is built to model the spatial and scale dependencies for feature fusion and reduce the layer-by-layer transmission of redundancy or interference information. Finally, a dual-discriminator-based DA (DDA) block is designed for class-specific semantic features and global structural alignments in both spatial and prediction spaces. It will enable HAMAE to model the cross-modal, cross-scale, and cross-domain associations, yielding more representative domain-invariant multimodal fusion features. Extensive experiments on five cross-domain MSRS datasets verify the superiority of the proposed HMDA-DANet over other state-of-the-art methods.**

***Index Terms*— Cross-scene, end-to-end classification, frequency attention, masked autoencoder (MAE), multiadversarial learning, multimodal masking learning, multisource remote sensing (MSRS) data.**

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## I. Introduction

With the development of remote sensing technologies, large-scale and high-quality Earth observation (EO) data have been captured. More comprehensive remote sensing data classification tasks and methods centered on hyperspectral images (HSIs) have been formed, gradually transitioning from single-source remote sensing data (i.e., HSIs) classification [1] to multisource remote sensing (MSRS) data [i.e., HSIs, light detection and ranging (LiDAR), synthetic aperture radar (SAR)] classification [2], [3], improving classification accuracy of land covers for various applications [4], [5], [6].

Traditional classification methods were designed based on the assumption of the same distribution, where the models trained on the labeled samples are applied to test other unlabeled samples in the same scene. Due to various factors such as sensor types, diurnal variations, seasonal changes, meteorological conditions [7], the practical environment often violates this assumption, resulting in spectral shifts [8] and spatial texture variations [9]. There are significant differences in the distributions of training data [source domain (SD)] and testing data [target domain (TD)]. Moreover, the acquisition of large-scale labeled MSRS data in practice is laborious and expensive [10]. These will result in poor performance of trained models in the TD. This phenomenon is called domain shift [11]. A natural idea is to utilize the labeled samples in the SD to assist in the classification of unlabeled samples in the TD, promoting the research on domain adaptation (DA) [12].

Existing DA methods can be divided into three categories: statistical feature transformation, geometric feature transformation, and deep learning [13]. Maximum mean discrepancy (MMD), as a common statistics-based method, was proposed to reduce the gaps by maximizing the mean difference between the SD and TD [14]. Substructural optimal transport was devised to use the local information of data for knowledge transfer [15]. Yu et al. [16] first proposed the idea

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of dynamic adversarial distribution adaptation for domain-invariant representation. Subsequently, the DA-based methods have been introduced into cross-domain HSI classification, such as topological structure and semantic information transfer network [17], joint adversarial network [18], [19], consistency-aware customized learning approach [20], achieving feature-, domain-, or class-level alignments. Although these methods improved the cross-domain performance, the special imaging mechanism of hyperspectral sensors makes it susceptible to some environmental factors, where the per-pixel solar and sky ancillary illumination information present significant differences, yielding poor performance using a single HSI in some practical scenarios [21]. As such, some researches have tried to explore the complementary information of MSRS data, where some auxiliary images that are robust to the domain variations are introduced to improve the cross-domain classification performance. There is an urgent need for analysis and interpretation of multidimensional, multimodal, and multidomain information. Accordingly, a new classification task, i.e., cross-domain classification with MSRS data, emerged in the remote sensing community. Zhao et al. [22] first designed a fractional fusion and spatial–spectral DA (FrF-SSDA) method, utilizing the stable elevation and spatial information from LiDAR data for spectral alignment of HSIs. Zhang et al. [23] defined the problem of cross-domain MSRS data classification, and proposed a multilevel DA network (MDA-NET) to improve the domain alignment of topological structures. In addition, there are also some models that attempt to introduce other images (i.e., SAR [24], RGB [25]) to overcome the domain shift of HSIs and improve the cross-domain classification accuracy.

However, the above methods usually relied mainly on high-quality labeled HSI or MSRS data, and failed to utilize a large amount of unlabeled data between domains, limiting their robustness and performance. In recent years, self-supervised learning has become a new paradigm to enable deep learning models to learn representative features from a large number of unlabeled samples without requiring extensive annotations [26]. Different from the contrastive learning [27], [28], masked image modeling (MIM) showed great advantages and potential in perceiving spatial details in images and learning fine-grained (FG) semantic representations, receiving widespread attention in the field of computer vision [29]. Compared to supervised methods, such self-supervised learning methods may be more suitable for HSI or MSRS classification. Although they continued to develop in the field of remote sensing [30], [31], there is still a lack of relevant research in the field of HSI classification, especially in the field of MSRS data classification. At present, the MIM-based research mainly faces the following issues.

1) Multiscale learning has always been one of the key factors to improve the classification performance of HSIs or MSRS data [32]. However, existing MIM-based methods were designed using traditional masked autoencoders (MAEs) [33], ignoring the roles of low-level representations. Moreover, they usually required the specialized pre-training processes, increasing their complexity and demand for pre-training data.
2) Existing algorithms utilized the small-sized local window as input to generate overlapping patches [34]. When applied to large-scale HSI or MSRS data, it is difficult for them to learn reliable long-range relationships of data within a limited window [35]. In addition, processing a large number of such patches requires a significant amount of computational overhead, which is not conducive to practical applications.
3) For cross-domain MSRS data classification, there is an urgent need for analysis and interpretation of data with multidimensional, multimodal, and multidomain information [23]. However, there is a lack of in-depth research on multimodal MIM-based methods for such a task.

Aiming at the above issues, a hierarchical masked dual-adversarial DA network (HMDA-DANet) is proposed for cross-domain end-to-end classification of MSRS data. The main contributions are summarized as follows.

1) A hierarchical asymmetric MAE (HAMAE) is designed to extract rich multimodal multiscale features from MSRS data. Particularly, a frequency dynamic large-scale convolutional (FDLConv) block is built to focus more on the structure of data to enhance important information, and an intramodality enhancement and intermodality interaction (IAEIEI) block is built for multimodal interaction and cross-modal reconstruction.
2) A multimodal multiscale feature fusion (MMFF) block is devised to fully model the long-range spatial and scale dependencies between multimodal multiscale features. Moreover, a dual-discriminator-based DA (DDA) block is further designed to align the class-specific semantic feature and global structural information, eliminating spectral shifts and distribution differences between cross-domain MSRS data.
3) An HMDA-DANet framework is built to carefully construct cross-modal, cross-scale, and cross-domain associations, extracting more discriminative domain-invariant features and improving cross-domain classification accuracy. Particularly, it is the first attempt to design a multimodal MAE that can be trained from scratch for multiscale feature learning and cross-domain classification of MSRS data.

The remainder of this article is organized as follows. Section II reviews the related work for cross-domain MSRS data classification and MIM. Detailed framework and optimization process of the proposed method are described in Section III. Section IV reports the experimental results, comparative analysis, and ablation study, followed by the conclusion in Section V.

## II. Related Work

### A. Cross-Domain MSRS Data Classification

In the past few years, deep learning has made great progress in cross-domain MSRS data classification. The cross-domain MSRS classification task was introduced in [22] and [23], where FrF-SSDA with adversarial DA, and MDA-NET with optimal transport and heterogeneous knowledge graphs

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matching are designed. Yang et al. [36] proposed a graph embedding interclass relation-aware adaptive network, where a joint de-scrambling alignment strategy and an adaptive inter-class topology are utilized to perceive inter-class relation. Based on sparse representation, a contrastive-constrained cross-scene model-informed interpretable classification strategy was devised, achieving model-interpretable decoupling of DA and classification tasks [13]. A shared-private decoupling-based multilevel feature alignment semisupervised (SASS) learning method was built to learn shared representations of the SD and TD and generate valuable pseudolabels for cross-domain classification [37]. The above methods are all patch-based with high complexity. To improve understanding of cross-city environments, IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS) organized a cross-city semantic segmentation contest (C2Seg) in 2023 to promote the development of semantic segmentation [38]. As such, Hong et al. [24] proposed a high-resolution DA network (HighDAN) with adversarial learning. Overcoming complex distribution differences with multimodal heterogeneity and transferring accurate supervision information are key issues. However, traditional adversarial DA only judged which domain the input comes from [18], [19], [22], [24], and [34], ignoring the transfer of class-related domain knowledge. In this article, we design a DDA block to eliminate multimodal distribution differences from both class-specific semantic and global structural alignments.

### B. Masked Image Modeling

As a generative self-supervised learning method, under the pretext of restoring masked pixels, MIM focuses on building some models that only utilize the visible parts to reconstruct inputs, adaptively learning their intrinsic representations by understanding the underlying structure of inputs. It is because of its unique learning advantages that it has made breakthroughs in the field of computer vision, and subsequently gained widespread attention in the field of remote sensing. Considering the spatial–spectral characteristics of HSIs, pixel-level/patch-level masking [39], spectral masking [40], spatial–spectral masking [41], and multiscale masking [42] methods were designed. Furthermore, to utilize the complementarity of MSRS data, by directly cascading them as input, Lin et al. [43] built a spatial–spectral MAE by performing spatial and spectral reconstructions for learning spatial–spectral representations; while Li et al. [44] used contrastive learning and MIM to explore structural representations for multimodal fusion. Du et al. [45] utilized the multilevel masking independently across different modalities for cross-modal interaction. Moreover, Fang et al. [34] combined self-distillation with spatial–spectral masking into DA to build a masked self-distillation DA (MSDA) framework for cross-domain HSI classification. In this article, instead of the above small-sized inputs and traditional MAE, we construct an HAMAE to fully explore rich multiscale representations and long-range multimodal interactions at low complexity for cross-domain end-to-end classification of MSRS data.

## III. Proposed HMDA-DANet Framework

### A. Architecture Overview

Our goal is to utilize the labeled SD data and unlabeled TD data for unsupervised DA (UDA) in end-to-end classification of MSRS data. Fig. 1 depicts the architecture of HMDA-DANet, with five components: preprocessing block (Section III-B), HAMAE encoder (Section III-C), HAMAE decoder (Section III-D), MMFF block and end-to-end classification head (Section III-E), and DDA block (Section III-F).

### B. Preprocessing Block

For notational clarity, we first introduce the symbols and definitions. Suppose that the MSRS data in the labeled SD are $\mathcal{S} = \{(\mathcal{X}_H^s, \mathcal{X}_L^s), Y^s\}$; while $\mathcal{T} = \{(\mathcal{X}_H^t, \mathcal{X}_L^t)\}$ in the unlabeled TD. $\mathcal{X}_H \in \mathbb{R}^{C_h \times W \times H}$ and $\mathcal{X}_L \in \mathbb{R}^{C_l \times W \times H}$ are the HSI and LiDAR/SAR data, respectively. $H$ and $W$ are the spatial size, $C_h$ and $C_l$ are the channel number. $Y^s$ is the known source label space with $N$ classes, while the target label space $Y^t$ is unknown. $s$ and $t$ are the SD and TD, respectively.

Take the SD as an example. Due to the inconsistent dimensions of MSRS data in different domains and considering the influence of their imaging mechanisms, noise, and other factors on the learning of discrimination information for land covers, a preprocessing block is designed in Stage I of Fig. 1. First, the 2-D convolutional (Conv) block with Conv layer, batch normalization (BN) layer, and rectified linear unit (ReLU) function is utilized to align their channel dimensions, whose outputs with size $C_0 \times W \times H$ can be described as

$$\begin{aligned} \mathcal{H}^s &= f_{2-\mathrm{D3}}^{C_0 \to C_0}\left(f_{2-\mathrm{D3}}^{C_h \to C_0}\left(\mathcal{X}_H^s\right)\right) \\ \mathcal{L}^s &= f_{2-\mathrm{D3}}^{C_l \to C_0}\left(\mathcal{X}_L^s\right) \end{aligned} \tag{1}$$

where $f_{2-\mathrm{D3}}^{* \to C_0}(\cdot)$ is the 2-D 3 × 3 Conv block with output channel number $C_0$ and stride 1. $*$ means the input channel number, such as $C_h$, $C_l$, and $C_0$.

Special imaging mechanism may introduce redundancy and interference to MSRS data. Particularly, the frequency domain-based feature learning methods can remove unimportant frequency components [46], and provide insights into image structures (i.e., edge, texture, shape) [47], [48] or generalizable features [49] in the fields of computer vision or remote sensing. Inspired by these frequency attentions, an FDLConv block is designed as a frequency domain filter. By dynamically modulating frequency components, it highlights important information and suppresses interference, providing clear and enhanced multimodal data for subsequent cross-modal reconstruction, more accurately exploring representative features from MSRS data. From Fig. 2, fast Fourier transform (FFT) is first used to transfer the inputs from the spatial domain to the frequency domain, whose results are fed into two large-scale 2-D depthwise Conv (2D-DWConv) layers. Then, the average pooling (AVP) and maximum pooling (MAP) layers are performed to model their global spatial relationship, followed by a 2-D 1 × 1 Conv layer and a Sigmoid function to yield two adaptive weights. Finally, inverse FFT (IFFT) is utilized to convert the outputs from the frequency domain to the spatial domain, followed by the residual connection

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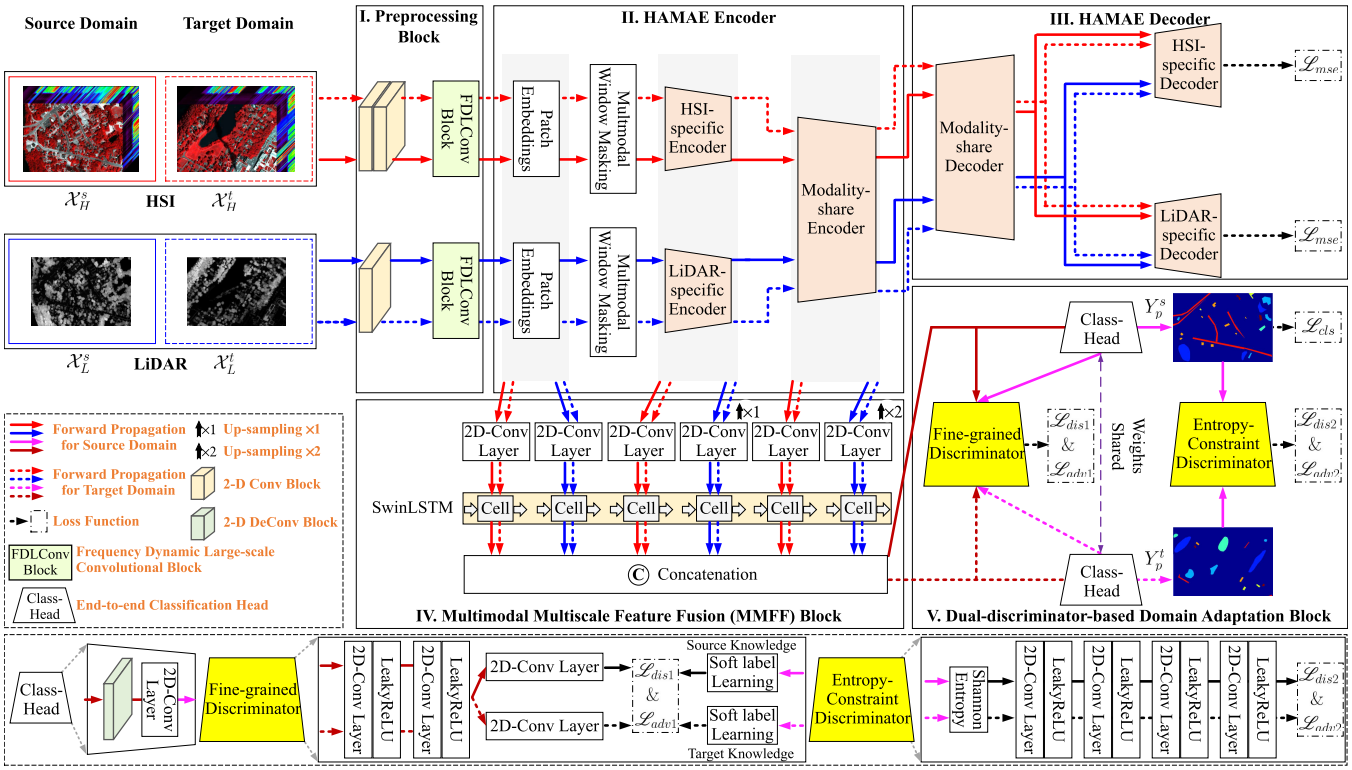



Fig. 1. Overall architecture of the proposed HMDA-DANet framework.

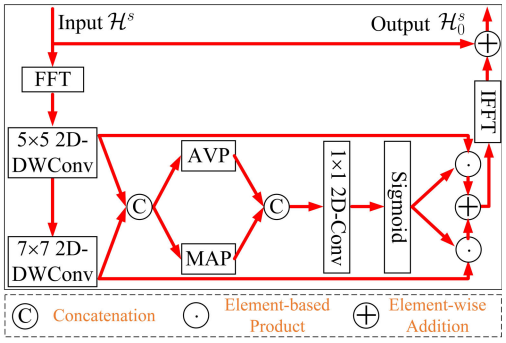



Fig. 2. Inner structure of FDLConv block.

to obtain the enhanced multimodal features. Take the HSI modality as an example, the enhanced features $\mathcal{H}_0^s$ with size $C_0 \times W \times H$ can be written as

$$\begin{aligned} T1, T2 &= f_{\text{dw5}}\left(\text{FFT}\left(\mathcal{H}^s\right)\right), f_{\text{dw7}}\left(f_{\text{dw5}}\left(\text{FFT}\left(\mathcal{H}^s\right)\right)\right) \\ [w1, w2] &= \sigma\left(f_{2-\text{D1}}^{2\to 2}\left(\left[f_{\text{avp}}([T1, T2]), f_{\text{map}}([T1, T2])\right]\right)\right) \\ \mathcal{H}_0^s &= \text{IFFT}(w1 \odot T1 \oplus w2 \odot T2) \oplus \mathcal{H}^s \end{aligned} \tag{2}$$

where $f_{\text{dw5}}(\cdot)$ and $f_{\text{dw7}}(\cdot)$ are the $5 \times 5$ and $7 \times 7$ 2D-DWConv layers, $T1, T2 \in \mathbb{R}^{C_0 \times W1 \times H}$, $w1, w2 \in \mathbb{R}^{1 \times W1 \times H}$, $\odot$ and $\oplus$ mean the element-based product and addition operations, respectively, $\sigma$ is the Sigmoid function, $f_{\text{avp}}(\cdot)$ and $f_{\text{map}}(\cdot)$ are the AVP and MAP layers, respectively, and $[\cdot, \cdot]$ is the concatenation operation. Similarly, the spatial-enhanced features for LiDAR/SAR modality are defined as $\mathcal{L}_0^s \in \mathbb{R}^{C_0 \times W \times H}$.

Moreover, based on the MAE, by using Swin transformer (SwinT) [50] as the backbone unit, a hierarchical asymmetric SwinT-MAE (HAMAE) is designed to extract rich multimodal multiscale features through multimodal masking learning and unsupervised training, as shown in Stages II and III of Fig. 1.

### C. HAMAE Encoder

In Fig. 3, the HAMAE encoder contains multimodal masking learning, two modality-specific encoders, and a modality-share encoder for multimodal multiscale feature extraction.

*1) Modality-Specific Encoder:* First, for HSI and LiDAR/SAR modalities, their spatial-enhanced features in (2) are partitioned into the patch formats using the patch embeddings module, generating some non-overlapping patch sets. Each patch contains $K \times K$ adjacent pixels, followed by a linear embedding layer to map the channel dimension from $C_0$ to $C_1$. Then, the window masking mechanism is introduced to design a multimodal masking learning strategy with a high masking ratio $\theta$ (i.e., the ratio of invisible patches, $\theta = 75\%$ [33] in all experiments) to replace the traditional masking method in [33], generating some masking features with only a small portion of visible patch sets to enable the whole model to learn more representative multimodal features from MSRS data.

Then, by utilizing the SwinT as the basic unit, two modality-specific encoders are developed to extract the rich multimodal features from the above masking features. From the left half of Fig. 3, two SwinT blocks are applied to learn the intramodality information, followed by the patch merging operation to reduce the spatial dimensions of multimodal features.

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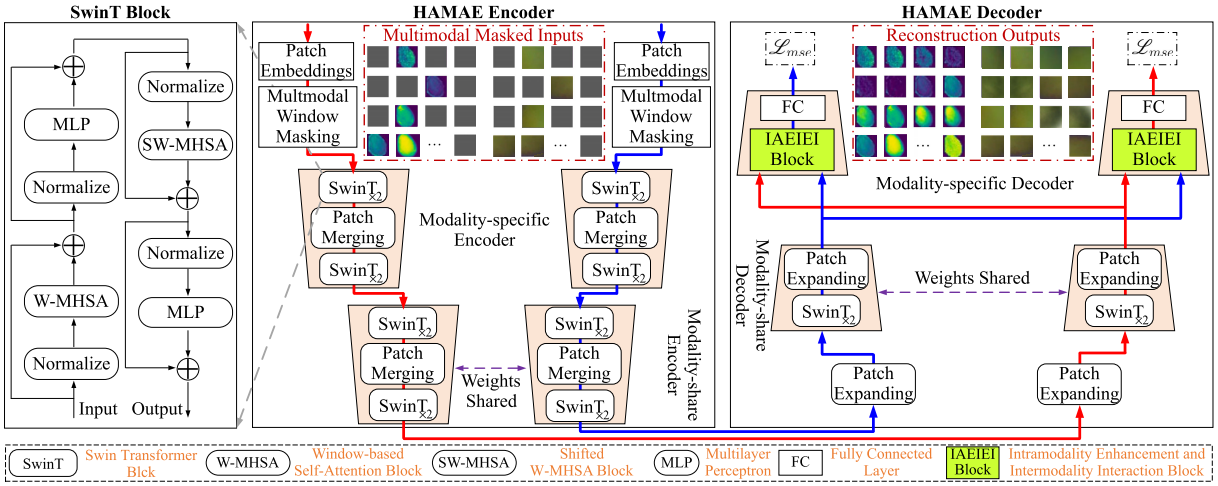



Fig. 3. Inner structure of the encoder and decoder in the designed HAMAE.

*2) Modality-Share Encoder:* To further model the inter-modality relationship, the modality-share encoder is further built. Its structure is the same as that of modality-specific encoder, which, however, is shared across different modalities.

Take the HSI modality as an example, the above process for the designed HAMAE encoder can be described as

$$\begin{aligned}\mathcal{H}_1^s &= f_{\text{mask}}\left(f_{\text{emb}}\left(\mathcal{H}_0^s\right)\right)\\ \mathcal{H}_2^s &= f_{\text{spe-en}}\left(\mathcal{H}_1^s\right)\\ \mathcal{H}_3^s &= f_{\text{sha-en}}\left(\mathcal{H}_2^s\right)\end{aligned} \tag{3}$$

where $f_{\text{emb}}(\cdot)$ is the patch embedding, $f_{\text{mask}}(\cdot)$ is the multimodal window masking, $f_{\text{spe-en}}(\cdot)$ is the modality-specific encoder, and $f_{\text{sha-en}}(\cdot)$ is the modality-share encoder. Hence, the multimodal multiscale features are obtained as $\mathcal{H}_i^s$ and $\mathcal{L}_i^s$ with size $2^{i-1}C_1 \times \lceil (W/K^i) \rceil \times \lceil (H/K^i) \rceil$ for the SD. $i = 1, 2, 3$, and $\lceil \cdot \rceil$ is the round up. Unlike natural images, due to the complexity and differences in the imaging mechanism and characteristics, $K$ is set to 2, and $C_1$ is 96 in all experiments to assist the whole model in extracting the domain-invariant features. Similarly, the multimodal multiscale features for the TD are $\mathcal{H}_i^t$ and $\mathcal{L}_i^t$.

### D. HAMAE Decoder

The HAMAE decoder contains a modality-share decoder and two modality-specific decoders for cross-modal reconstruction, as shown in the right half of Fig. 3. Different from the traditional MAE [33], since the mask tokens are not removed in our HAMAE encoder, there is no need to add the mask tokens in the HAMAE decoder.

*1) Modality-Share Decoder:* First, a simple modality-share decoder is built, where two SwinT blocks and two patch expanding operations are executed alternately for upsampling twice to recover the spatial dimensions. Suppose that this decoder is defined as $f_{\text{sha-de}}(\cdot)$, and the preliminarily reconstructed multimodal features from the masked multimodal data for HSI and LiDAR/SAR modalities can be written as $\mathcal{H}_1^{s\prime} = f_{\text{sha-de}}(\mathcal{H}_3^s)$ and $\mathcal{L}_1^{s\prime} = f_{\text{sha-de}}(\mathcal{L}_3^s)$, respectively.

*2) Modality-Specific Decoder:* To achieve more accurate cross-modal reconstruction, a modality-specific decoder is further designed, containing two IAEIEI blocks with linear layers. Particularly, traditional self- or cross-attention blocks usually directly use the multiply projections of data themselves for modality feature enhancement or multimodal information interaction [3], [13], [25], [28], [34], [37], [40], [43], [44]. Despite achieving relatively good results, some important information within the modality or prior relationships between modalities may not be fully utilized, especially for the reconstruction task with high masking ratio. As such, the IAEIEI block is designed, as shown in Fig. 4, where some additional learnable vectors are introduced to build a memory-augmented self-attention subblock marked with purple dashed box and a memory-augmented cross-attention subblock marked with green dashed box. On the one hand, more priori knowledge about intramodality and intermodality information may be introduced to improve the cross-modal reconstruction quality, even under high masking rate. On the other hand, these additional vectors can expand the reconstruction space to some extent, which will embed some additional information beyond the SD to the extracted multimodal hierarchical features, thus improving their generalization to the TD.

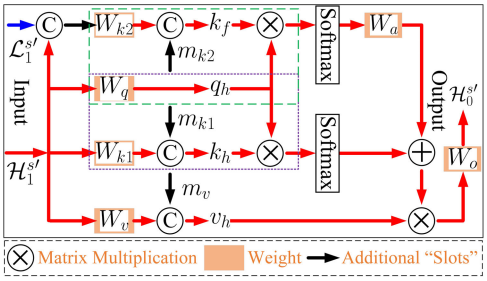



Fig. 4. Inner structure of IAEIEI block.

For the memory-augmented self-attention subblock, the preliminarily reconstructed features $\mathcal{H}_1^{s\prime}$ of HSI modality are projected as the intramodality query $q_h$, key, and value with projection matrixes (i.e., $W_q$, $W_{k1}$, $W_v$) using three fully connected (FC) layers. Then, an additional key $m_{k1} \in \mathbb{R}^{n_m \times (h \times d)}$ and an additional value $m_v \in \mathbb{R}^{n_m \times (h \times d)}$ are defined as the learnable vectors, which are cascaded with the original key and value to obtain the intramodality enhancement key $k_h$

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and value $v_h$. $n_m$ is the additional token number, $h$ and $d$ are the head number and dimension in the multihead attention, respectively. $k_h$ is multiplied with $q_h$, followed by a Softmax function to obtain the self-attention weight $A_{\text{sa}}$ to model the intramodality relationship. For the designed memory-augmented cross-attention subblock, to integrate the multimodal complementarity to accurately reconstruct the HSI modality, the preliminarily reconstructed multimodal features $\mathcal{H}_1^{s\prime}$ and $\mathcal{L}_1^{s\prime}$ are first cascaded together, whose results are mapped to the intermodality key by an FC layer with matrix $W_{k2}$. Additional key $m_{k2}$ is further introduced to obtain the intermodality enhancement key $k_f$, which is multiplied with $q_h$, followed by a Softmax function and an FC layer with matrix $W_p$ to obtain the cross-attention weight $A_{\text{ca}}$ to effectively model the intermodality relationship.

Finally, the self-attention weight $A_{\text{sa}}$ and cross-attention weight $A_{\text{ca}}$ are fused, whose result is multiplied with value $v_h$. An FC layer with projection matrix $W_o$ is further adopted to generate the refined reconstructed feature $\mathcal{H}_0^{s\prime} \in \mathbb{R}^{C_0 \times W \times H}$ with the IAEIEI for HSI modality, as follows:

$$\begin{aligned} q_h, k_h, v_h &= f_q\left(\mathcal{H}_1^{s\prime}\right), \quad \left[f_{k1}\left(\mathcal{H}_1^{s\prime}\right), m_{k1}\right], \ \left[f_v\left(\mathcal{H}_1^{s\prime}\right), m_v\right] \\ A_{\text{sa}} &= _{softmax}\left(q_h \otimes k_h\right) \\ k_f &= \left[f_{k2}\left(\left[\mathcal{H}_1^{s\prime}, \mathcal{L}_1^{s\prime}\right]\right), m_{k2}\right] \\ A_{\text{ca}} &= f_p\left(_{softmax}\left(q_h \otimes k_f\right)\right) \\ \mathcal{H}_0^{s\prime} &= f_o\left(\left(A_{\text{sa}} \oplus A_{\text{ca}}\right) \otimes v_h\right) \end{aligned} \tag{4}$$

where $f_q(\cdot)$, $f_{k1}(\cdot)$, $f_{k2}(\cdot)$, $f_v(\cdot)$, $f_p(\cdot)$, and $f_o(\cdot)$ are FC with projection weights $W_q$, $W_{k1}$, $W_{k2}$, $W_v$, $W_p$, and $W_o$, respectively. $\otimes$ is the matrix multiplication. Similarly, the refined reconstructed features for LiDAR/SAR modality are defined as $\mathcal{L}_0^{s\prime}$.

Through multimodal masking learning and unsupervised training in the HAMAE structure, using only a small portion of visible patches from the input multimodal data can effectively extract the rich and representative multimodal multiscale features $\mathcal{H}_i^s$, $\mathcal{L}_i^s$, and the corresponding reconstructed features $\mathcal{H}_0^{s\prime}$, $\mathcal{L}_0^{s\prime}$ in the SD, respectively. In the same way, as for the TD, the extracted multimodal multiscale features are obtained as $\mathcal{H}_i^t$, $\mathcal{L}_i^t$, and the corresponding reconstructed features are $\mathcal{H}_0^{t\prime}$, $\mathcal{L}_0^{t\prime}$. These features will be applied for subsequent cross-domain end-to-end classification of MSRS data.

### *E. MMFF and Classification*

After obtaining the representative multimodal multiscale features from MSRS data in the SD and TD, inspired by the SwinLSTM module, which can jointly model the effective spatiotemporal dependencies [51], a SwinLSTM-based MMFF and classification block is further built to fuse the multimodal multiscale features for end-to-end classification, which contains an MMFF block in Stage IV and a classification head in Stage V of Fig. 1.

First, the HAMAE encoder is reused to extract multimodal multiscale features $\mathcal{H}_i^s$, $\mathcal{L}_i^s$ for the SD, and $\mathcal{H}_i^t$, $\mathcal{L}_i^t$ for the TD from all patches of inputs. Then, in the MMFF block, these features are, respectively, fed into six 2-D $1 \times 1$ Conv layers and four upsample layers to align their channel and spatial dimensions, where BN and ReLU are not used. These aligned features are further listed as a feature sequence, and the SwinLSTM block is introduced by using the SwinLSTM cell as a basic unit to model their long-range spatial and scale dependencies. Finally, the output features are cascaded along the channel dimension to generate the multimodal multiscale fusion features $\mathcal{F}^s$ with size $6C_f \times \lceil (W/K) \rceil \times \lceil (H/K) \rceil$ for the SD as

$$\begin{aligned} \mathcal{H}_i^s &= f_{2-\text{D1}}^{C_i \to C_f}\left(\mathcal{H}_i^s\right), \quad \mathcal{L}_i^s = f_{2-\text{D1}}^{C_i \to C_f}\left(\mathcal{L}_i^s\right), \quad i = 1, 2, 3 \\ \mathcal{H}_j^s &= f_{\text{up}}\left(\mathcal{H}_j^s, j-1\right), \quad \mathcal{L}_j^s = f_{\text{up}}\left(\mathcal{L}_j^s, j-1\right), \quad j = 2, 3 \\ \mathcal{F}^s &= \text{SwinLSTM}\left(\left\{\mathcal{H}_1^s, \mathcal{L}_1^s, \mathcal{H}_2^s, \mathcal{L}_2^s, \mathcal{H}_3^s, \mathcal{L}_3^s\right\}\right) \end{aligned} \tag{5}$$

where $f_{2-\text{D1}}^{C_i \to C_f}(\cdot)$ is the 2-D $1 \times 1$ Conv layer with output channel $C_f$, $f_{\text{up}}(\cdot, j-1)$ is the $(j-1)\times$ upsample with bilinear interpolation, and $\{\cdot, \cdot\}$ is the sequence. For convenience, the above processes are defined as $\mathcal{F}^s = G(\mathcal{X}_H^s, \mathcal{X}_L^s)$ for the SD and $\mathcal{F}^t = G(\mathcal{X}_H^t, \mathcal{X}_L^t)$ for the TD, where $G$ is the generator.

After that, an end-to-end classification head that is defined as classifier $C$ is built to map the above features from the feature space to class space, containing some deconvolutional (DeConv) and Conv layers to predict the final classification map $Y_p^s$ for the SD, which can be described as

$$Y_p^s = C\left(\mathcal{F}^s\right) = f_{2-\text{D1}}^{128 \to N}\left(f_{D2D2}^{6C_f \to 128}\left(\mathcal{F}^s\right)\right) \tag{6}$$

where $f_{D2D2}^{6C_f \to 128}(\cdot)$ is the 2-D $2 \times 2$ DeConv layer with output channel 128 and stride 2, and $f_{2-\text{D1}}^{128 \to N}(\cdot)$ is the 2-D $1 \times 1$ Conv layer with output channel $N$ and stride 1. The number of 2-D DeConv layers is $\lceil (K/2) \rceil$.

Finally, through the generator $G$ and the classifier $C$, the cross-entropy loss $\mathscr{L}_{\text{ce}}$ and the dice loss $\mathscr{L}_{\text{di}}$ are utilized to build the classification loss $\mathscr{L}_{\text{cls}}$ for end-to-end classification and alleviating the class imbalance issue caused by certain factors across domains, which can be represented as

$$\begin{aligned} \mathscr{L}_{\text{cls}} = \mathscr{L}_{\text{ce}}\left(Y^s, Y_p^s\right) + \mathscr{L}_{\text{di}}\left(Y^s, Y_p^s\right) \\ + \mathscr{L}_{\text{mse}}\left(\mathcal{H}_0^s, \mathcal{H}_0^{s\prime}\right) + \mathscr{L}_{\text{mse}}\left(\mathcal{L}_0^s, \mathcal{L}_0^{s\prime}\right) \end{aligned} \tag{7}$$

where $\mathscr{L}_{\text{mse}}(\cdot, \cdot)$ refers to the mean-squared error loss between the invisible patches and the reconstructed ones.

In the same way, the predicted classification map $Y_p^t$ can be obtained for the TD. However, due to differences in lighting conditions, climate environment, and sensor characteristics, there are significant spectral drift and distribution differences in the MSRS data between training and testing scenarios. It is this negative shift that causes the classification model trained in the SD to fail to obtain satisfactory prediction results in the TD. Therefore, it is necessary to design an effective DA strategy to eliminate domain shift to improve the classification accuracy of the TD.

Unlike existing hierarchical architectures (i.e., U-shaped structures) and feature fusion methods (i.e., concatenation, summation), by modeling the long-range spatial and scale dependencies of multimodal hierarchical features, our HAMAE and MMFF block not only explore hierarchical correlations from low to deep levels but also transfer the multimodal detail information of low-level features to high-level features, effectively reducing the layer-by-layer transmission

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of redundancy or interference information and improving the efficiency and performance of cross-modal and cross-scale feature fusion.

### F. Dual-Discriminator-Based DA

After multimodal feature extraction and fusion, inspired by adversarial learning [52], [53], a DDA block is designed to overcome the domain shifts for cross-domain classification of MSRS data. As shown in Stage V of Fig. 1, it contains an FG discriminator $D_1$ and an entropy-constraint (EC) discriminator $D_2$ to align the semantic feature and global structural information in each class between domains, respectively.

Traditional discriminator only judges which domain the input comes from, until it cannot distinguish between the SD and TD [18], [19], [22], [24], [34]. However, on the one hand, the MSRS data present different imaging characteristics and complex multimodal heterogeneity, which are easily affected by some factors (i.e., sensor differences, environment changes), forming larger domain gaps. On the other hand, even if the scene changes are the same, their impacts on the land cover vary with the differences in classes. For example, class "Trees" exhibits completely various attributes in different seasons or climates, while class "Buildings" does not.

As such, first, an FG discriminator $D_1$ with multichannel soft label learning is built for class-level alignment on the multimodal fusion features in (5), eliminating domain gaps of each class in the feature space. As shown in the bottom row of Fig. 1, instead of utilizing the binary domain encodings (i.e., [0, 1] for the SD, [1, 0] for the TD), inspired by [54], the multichannel soft labels are considered as the domain encodings, which is defined as $[0, E^s]$ for the SD, while $[E^t, 0]$ is for the TD, respectively. $E^s$ and $E^t$ mean the $N$-dimensional soft domain encoding with the same size as the label $Y^s$, which can be calculated as

$$E_n^s = \frac{e^{(Y_p^s)^n/T}}{\sum_{i=1}^{N} e^{(Y_p^s)^i/T}}, \quad E_n^t = \frac{e^{(Y_p^t)^n/T}}{\sum_{i=1}^{N} e^{(Y_p^t)^i/T}}$$
$$E^s = \begin{cases} E^s > \theta, & \theta \\ \text{others}, & E^s, \end{cases} \quad E^t = \begin{cases} E^t > \theta, & \theta \\ \text{others}, & E^t \end{cases} \tag{8}$$

where $T$ is the temperature coefficient with fixed value 1.8 to smooth the probability distribution, and $\theta$ is the confidence threshold with fixed value 0.9, $n = 1, 2, \ldots, N$. Then, $D_1$ uses two 2-D $3 \times 3$ Conv layers with LeakyReLU to compress the channels of multimodal fusion features, and another two 2-D $3 \times 3$ Conv layers and an upsample layer to map their output channel number to $N$ and restore the spatial resolution. Then, we obtain the source semantic feature $D_1(F^s)^{(0)}$ and the target semantic feature $D_1(F^t)^{(1)}$ for each class, respectively. The channel numbers of four Conv layers are 128, 64, $N$, and $N$, respectively. The discriminator loss $\mathscr{L}_{\text{dis1}}$ is defined as

$$\mathscr{L}_{\text{dis1}} = -\sum_{u,v}^{W,H}\sum_{n=1}^{N} E_n^{s(u,v)} \log\left(D_1\left(F^s\right)^{(n,u,v,0)}\right) - \sum_{u,v}^{W,H}\sum_{n=1}^{N} E_n^{t(u,v)} \log\left(D_1\left(F^t\right)^{(n,u,v,1)}\right) \tag{9}$$

and the corresponding adversarial loss $\mathscr{L}_{\text{adv1}}$ is written as

$$\mathscr{L}_{\text{adv1}} = -\sum_{u,v}^{W,H}\sum_{n=1}^{N} E_n^{t(u,v)} \log\left(D_1\left(F^t\right)^{(n,u,v,0)}\right). \tag{10}$$

Furthermore, to effectively prevent the overfitting problem and improve the generalization of the whole HMDA-DANet framework, an EC discriminator $D_2$ is developed to explore the structural consistency between the SD and TD by enforcing the entropy maps from the target predictions to be similar to the source ones. Because the obtained entropy map encodes the global information of all classes, the EC discriminator is beneficial for driving the entire model to learn the cross-domain variation of global information to further eliminate the domain gaps, improving the prediction accuracy in the TD. As shown in the bottom row of Fig. 1, we first calculate Shannon entropy [55] based on the predicted results at pixel level to obtain the entropy maps for the SD and TD. After that, the adversarial learning strategy is introduced to perform the entropy-level DA, where the discriminator loss $\mathscr{L}_{\text{dis2}}$ can be calculated as

$$\mathscr{L}_{\text{dis2}} = \mathscr{L}_{\text{bce}}\left(D_2\left(H\left(Y_p^s\right)\right), 0\right) + \mathscr{L}_{\text{bce}}\left(D_2\left(H\left(Y_p^t\right)\right), 1\right) \tag{11}$$

where $H(\cdot)$ means Shannon entropy, $\mathscr{L}_{\text{bce}}(\cdot)$ is the binary cross-entropy with logits loss function, and the corresponding adversarial loss $\mathscr{L}_{\text{adv2}}$ is represented as

$$\mathscr{L}_{\text{adv2}} = \mathscr{L}_{\text{bce}}\left(D_2\left(H\left(Y_p^t\right)\right), 0\right). \tag{12}$$

By integrating the above two discriminators, the total discriminator loss $\mathscr{L}_{\text{dis}}$ and the total adversarial loss $\mathscr{L}_{\text{adv}}$ in the designed DDA block can be obtained as

$$\mathscr{L}_{\text{dis}} = \mathscr{L}_{\text{dis1}} + \lambda_1 \mathscr{L}_{\text{dis2}}$$
$$\mathscr{L}_{\text{adv}} = \mathscr{L}_{\text{adv1}} + \lambda_2 \mathscr{L}_{\text{adv2}} \tag{13}$$

where $\lambda_1$ and $\lambda_2$ are weighting factors with fixed values of 0.1.

### G. Network Optimization

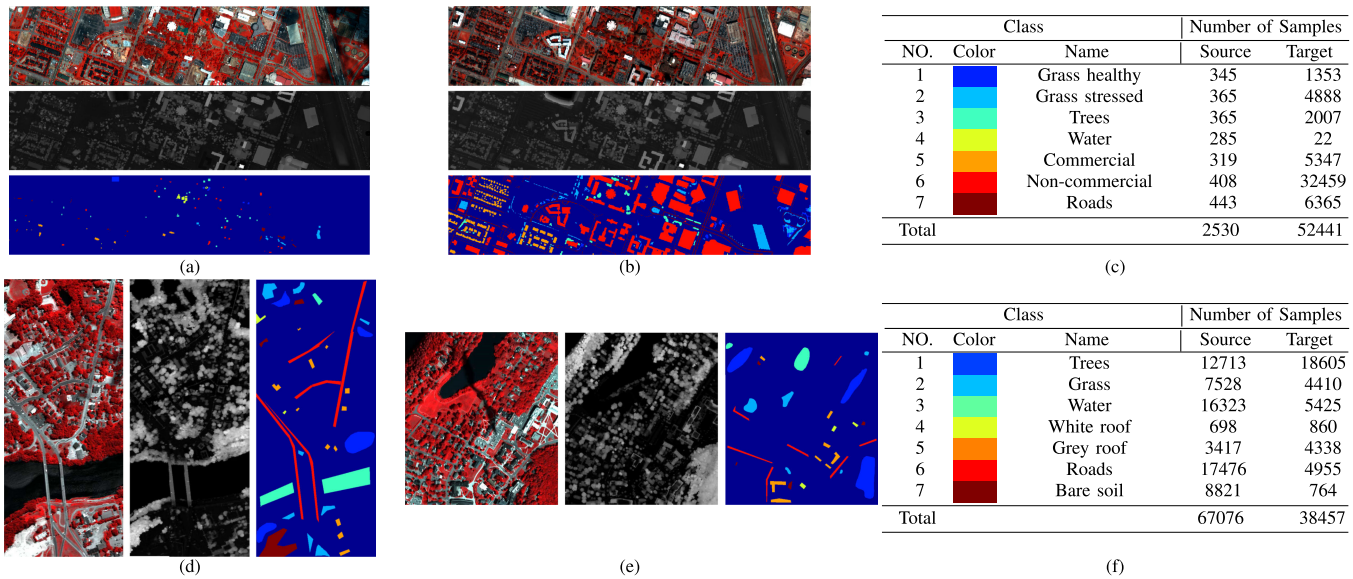
Based on the above designs, the overall min-max optimization problem in the HMDA-DANet framework is defined as

$$\arg\min_{G,C} \arg\max_{D_1,D_2} \mathscr{L}_{\text{cls}} + \mathscr{L}_{\text{adv}} - \mathscr{L}_{\text{dis}}. \tag{14}$$

During training, by alternatively optimizing the generator $G$, classifier $C$, and discriminators $D_1$ and $D_2$, the multimodal features from the SD and TD are aligned in a more FG strategy, improving the cross-domain classification accuracy.

Different from existing UDA methods [13], [22], [23], [24], [36], [37], by integrating the advantages of self-supervised learning and a patch-free end-to-end framework, the proposed HMDA-DANet fully utilizes the valuable unlabeled samples that have not been effectively utilized to assist the training process. This not only reduces its dependence on labeled samples to a certain extent but also effectively lowers the complexity, thereby achieving better cross-domain classification performance of MSRS data, while maintaining more competitive inference efficiency. Particularly, unlike existing MAE-based methods [25], [29], [30], [31], [32], [33], [34],

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(a) (b)

| Class | | | Number of Samples | |
|---|---|---|---|---|
| NO. | Color | Name | Source | Target |
| 1 | | Grass healthy | 345 | 1353 |
| 2 | | Grass stressed | 365 | 4888 |
| 3 | | Trees | 365 | 2007 |
| 4 | | Water | 285 | 22 |
| 5 | | Commercial | 319 | 5347 |
| 6 | | Non-commercial | 408 | 32459 |
| 7 | | Roads | 443 | 6365 |
| Total | | | 2530 | 52441 |

(c)

(d) (e)

| Class | | | Number of Samples | |
|---|---|---|---|---|
| NO. | Color | Name | Source | Target |
| 1 | | Trees | 12713 | 18605 |
| 2 | | Grass | 7528 | 4410 |
| 3 | | Water | 16323 | 5425 |
| 4 | | White roof | 698 | 860 |
| 5 | | Grey roof | 3417 | 4338 |
| 6 | | Roads | 17476 | 4955 |
| 7 | | Bare soil | 8821 | 764 |
| Total | | | 67076 | 38457 |

(f)

Fig. 5. (Top or left) False-color map of HSI data. (Middle) Gray-scale map of LiDAR data. (Bottom or right) Ground-truth map. (a)–(c) H13-H18 dataset. (d)–(f) NA-HA dataset.

[39], [40], [41], [42], [43], [44], [45], this is the first attempt to build a multimodal MAE that is trained from scratch for cross-domain end-to-end classification of MSRS data.

## IV. Experimental Results

To effectively validate the classification performance of the proposed HMDA-DANet framework for cross-domain end-to-end classification of MSRS data, some DA methods, e.g., SCLUDA [27], MLUDA [28], MSDA [34], FrF-SSDA [22], MDA-NET [23], HighDAN [24], and SASS [37], are selected as the comparison methods. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient ($\kappa$) are the selected quantitative metrics, and the average values of each metric with five Monte Carlo runs are reported in the following experiments. In addition, our HMDA-DANet is built with the Pytorch platform (i.e., Pytorch 2.0.0, python 3.10.12), and trained on a personal computer with an Intel Core i5-13600KF processor and an Nvidia GeForce RTX 4070 GPU.

### A. Experimental Data

The Houston13-Houston18 (H13-H18), Nashua-Hanover (Na-Ha), Houston13-Trento (H13-TO), Trento-Muffl (TO-MU), and Augsburg-Berlin (C2Seg-AB) datasets are selected as the cross-domain MSRS datasets, as shown in Figs. 5 and 6.

*1) H13-H18 Dataset:* It contains two city scenes with HSI and LiDAR data, i.e., Houston13 (SD) and Houston18 (TD), captured by the CASI-1500 sensor over the University of Houston campus and its neighboring area, Houston, TX, USA. Houston13 data were released by the IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest in 2013[1], while Houston18 data were released in 2018.[2] Their overlap area is reserved for analysis, with seven consistent land covers and 48 co-occurrence spectral bands for HSIs. There are $210 \times 954$ pixels in the SD and TD; however, their spatial and spectral resolutions are different. This dataset can be applied for cross-temporal analysis over the same city scenes.

*2) Na-Ha Dataset:* It is composed of two city scenes, i.e., Nashua (SD) and Hanover (TD), collected over the cities of Nashua and Hanover, USA, in 2014, respectively, which come from G-LiHT data.[3] Among them, Nashua data provide the co-registered LiDAR and HSI with $450 \times 1050$ pixels and a ground sample distance (GSD) of 1 m, while Hanover data provide the co-registered LiDAR and HSI with $720 \times 620$ pixels at a GSD of 1 m. There are 114 spectral bands ranging from 0.42 to 0.95 $\mu$m for HSIs. In addition, this dataset contains seven classes, which can be applied for cross-city analysis.

*3) H13-TO Dataset:* From [22], Houston13 data were served as the SD. Trento data (TD) with co-registered LiDAR and HSI were captured by the AISA Eagle and Optech ALTM 3100EA sensors over a rural area in Trento, Italy. There are $600 \times 166$ pixels with a spatial resolution of 1 m and 63 bands ranging from 420.89 to 989.09 nm. 63 co-occurrence bands and four classes are extracted for cross-temporal and cross-area analysis.

*4) TO-MU Dataset:* According to [36], Trento data were the SD, while Muffl data (TD) were collected by the ITRES CASI-1500 sensor over the University of Southern Mississippi's Gulf Park Campus, Long Beach, Mississippi, USA, in 2010,[4] with co-registered HSI and LiDAR. This dataset consists of $325 \times 220$ pixels with spatial resolution 1 m, 64 bands ranging from 375 to 1050 nm, and three consistent land covers for cross-sensor and cross-temporal analysis over the rural and university areas.

[1][Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2][Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion/2018-ieee-grss-data-fusion-contest/

[3][Online]. Available: https://glihtdata.gsfc.nasa.gov

[4][Online]. Available: https://github.com/GatorSense/MUUFLGulfport

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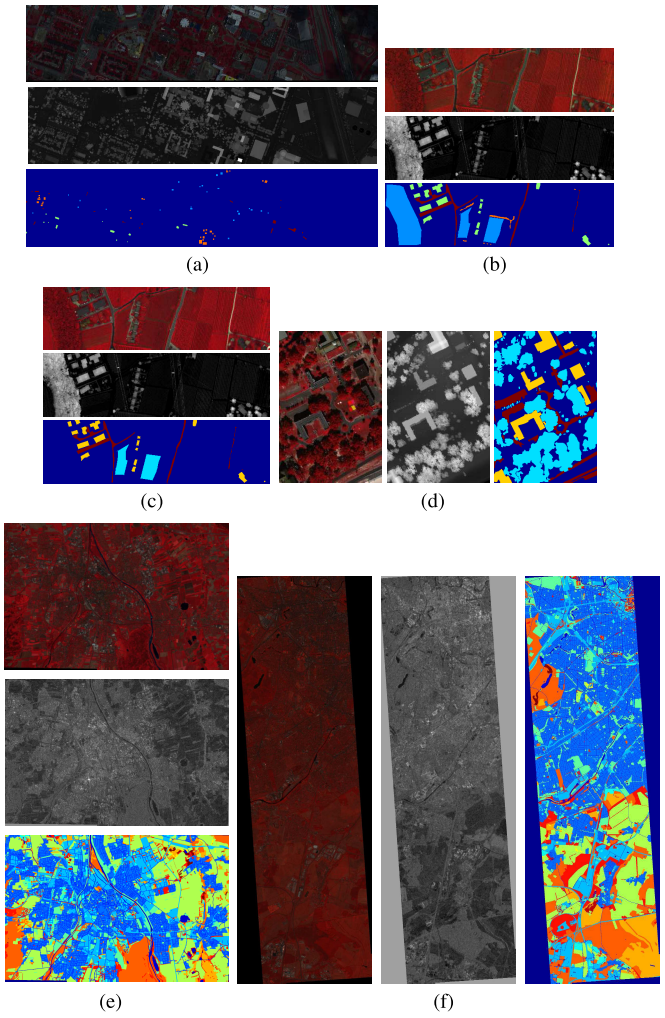



Fig. 6. (Top or left) False-color map of HSI data. (Middle) Gray-scale map of LiDAR or SAR data. (Bottom or right) Ground-truth map. (a) and (b) H13-TO dataset. (c) and (d) TO-MU dataset. (e) and (f) C2Seg-AB dataset.

*5) C2Seg-AB Dataset:* It was released by IEEE WHISPERS C2Seg Challenge in 2023[5] [24], [38], providing the co-registered HSI, MSI, and SAR data with a GSD of 10 m and 13 distinct land covers. For the HSI component from EnMAP satellite, Augsburg data (SD) contain 886 × 1360, and Berlin data (TD) contain 2465 × 811 pixels. They share the same bands (i.e., 242) in the wavelength range of 400–2500 nm. The SAR component is collected by Sentinel-1, with dual-polarized information. The MSI component is not considered in our experiment. This dataset can be applied for cross-city analysis. Different from traditional patch-based methods, the patch-free method [24], [35] is adopted in our HMDA-DANet framework. By cropping the whole scenes with the sliding window at certain intervals, we collect 174 and 60 images with a fixed size of 64 × 64 pixels as the SD training samples and the TD testing samples on the H13-H18 dataset, 448 and 110 images on the NA-HA dataset, 174 and 30 images on the H13-TO dataset, 90 and 24 images on the TO-MU dataset, and 1134 and 507 images on the C2Seg-AB dataset, respectively.

[5][Online]. Available: https://github.com/danfenghong/RSE_Cross-city

## B. Experimental Settings

To verify the classification performance of the HMDA-DANet framework, three groups of experiments are conducted on the five cross-domain MSRS datasets.

*1) Parameter Setting Analysis:* From Fig. 1, the first experiment is designed to analyze some key parameters in our HMDA-DANet, i.e., number $C_0$, number $C_f$, window size $w_s$, and token number $n_m$. In addition, their candidate sets are defined as $C_0 \in \{32, 64, 128\}$, $C_f \in \{32, 64, 96, 128\}$, $w_s \in \{1, 2, 4, 8\}$, and $n_m \in \{1, 3, 5, 7\}$, respectively.

*2) Classification Performance Analysis:* The second experiment is conducted for comparison with the literature, where SCLUDA [27], MLUDA [28], MSDA [34], FrF-SSDA [22], MDA-NET [23], HighDAN [24], and SASS [37] are used as the comparison methods. In addition, the efficiency of all considered methods is also analyzed and compared, where the average testing time (s) is reported.

*3) Ablation Study:* Based on the framework in Fig. 1, there are some core blocks designed for domain-invariant multimodal feature extraction and fusion, such as the FDLConv, IAEIEI, MMFF, and DDA blocks. The third experiment is further designed to validate the contribution of each component on the cross-domain end-to-end classification performance of MSRS data.

## C. Parameter Setting Analysis

From Section IV-B, there are four key parameters that need to be tuned in the proposed HMDA-DANet framework. For the convenience of analysis and practical application, their impacts on cross-domain classification performance are first measured on the H13-H18 and NA-HA datasets, as shown in Fig. 7. And then, these parameters are fixed and further applied to the H13-TO, TO-MU, and C2Seg-AB datasets.

*1) Analysis of Number $C_0$:* $C_0$ is the first 2-D Conv layer to align the channel number of input multimodal data, whose effects on cross-domain classification performance are shown in Fig. 7(a). As the value of $C_0$ increases within a certain range, the classification performance gradually improves. However, as its value continues to increase, the complexity and optimization difficulty of the whole model increase, which in turn affects the classification performance. Therefore, when the value of $C_0$ is set to 64, our HMDA-DANet framework reaches the quasi-optimal performance.

*2) Analysis of Number $C_f$:* It is the channel number in the MMFF block. Fig. 7(b) illustrates its influence on the cross-domain classification accuracy, where the quasi-optimal value of $C_f$ is 64.

*3) Analysis of Window Size $w_s$:* In the HAMAE structure and the MMFF block, $w_s$ is the window size of the window attention and shift window attention. As shown in Fig. 7(c), as the value of $w_s$ increases to 8, the classification performance of the proposed HMDA-DANet framework improves and then deteriorates gradually. It achieves the promising performance when $w_s$ is fixed to 4.

*4) Analysis of Number $n_m$:* It is the number of additional memory prior in the IAEIEI block to fully model the intramodality and intermodality relationships of MSRS data.

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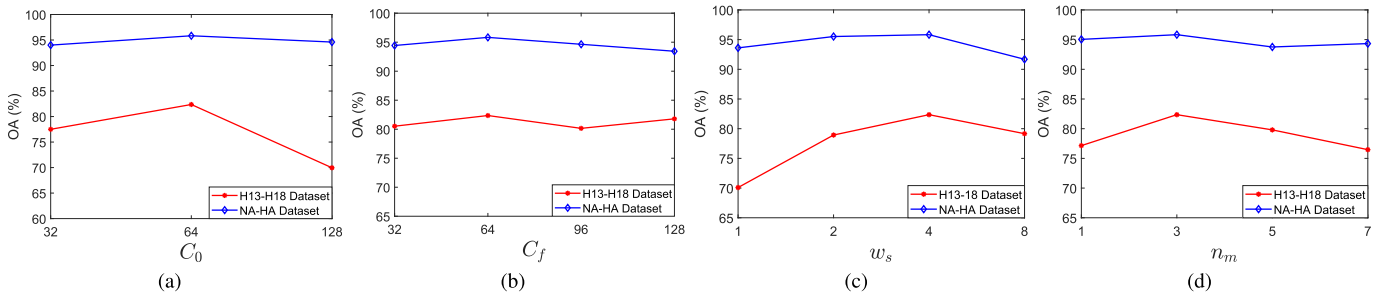



Fig. 7. OA (%) achieved by the proposed HMDA-DANet framework with different parameters for the H13-H18 and NA-HA datasets. (a) Number $C_0$ for multimodal channel alignment. (b) Number $C_f$ for multiscale channel alignment. (c) Window size $w_s$. (d) Number $n_m$ of memory tokens.

Fig. 7(d) reports the effects of $n_m$ on classification performance of HMDA-DANet. The larger the value of $n_m$ within a certain range, the higher the cross-domain classification accuracy. However, as its value continues to increase, more interference may be introduced. Therefore, the OA value of HMDA-DANet reaches its maximum value when $n_m$ is set to 3.

Particularly, the values of all parameters in our HMDA-DANet framework are exactly the same except for the learning rate, where the initial learning rate is 0.0005, 0.0001, 0.001, 0.0001, and 0.0005 with the polynomial learning rate schedule for the five datasets, respectively. In addition, the batch size is set to 4 due to the memory limitations. More detailed parameter settings and analysis are shown in Fig. 7.

### D. Classification Performance Analysis

Tables I–V report the classification results (i.e., class-specific accuracy, OA, AA, and $\kappa$) of the proposed HMDA-DANet framework and other comparison methods under the H13-H18, NA-HA, H13-TO, TO-MU, and C2Seg-AB datasets, whose classification maps are shown in Figs. 8–12, respectively.

From the perspective of structure designs, SCLUDA combines the contrastive learning and confidence learning to enhance the separability of data within domain and the discriminability for the TD. MLUDA applies the multilevel alignment with contrastive learning to increase the inter-class distance between the SD and TD. MSDA introduces the class-separable adversarial training to prevent misclassification, and utilizes the masked self-distillation to extract the class-relevant knowledge for the TD. The above three methods directly use the cascaded MSRS data as input, limiting their performance on most cross-scene MSRS datasets due to the lack of multimodal designs. The adversarial learning and fractional fusion are jointly used to build the FrF-SSDA, where LiDAR data are applied to reduce the spectral shift of HSIs for cross-domain classification. MDA-Net uses the optimal transport constrained dynamic normalization alignment for HSIs and the heterogeneous knowledge graphs matching alignment for LiDAR data, achieving better classification accuracy. SASS designs the shared-private feature decoupling module to learn the shared features between domains, and the class-balanced pseudo-label generation strategy to exploit the unlabeled samples, overcoming the overfitting. HighDAN introduces the adversarial learning and high-resolution network to explore the high-to-low-resolution representations for cross-city semantic segmentation. However, the first six DA methods are all patch-based, whose abilities to extract the representative cross-domain cross-modal features are limited for lack of global information and high computational complexity. Although HighDAN is a patch-free method, there is a lack of specialized designs for HSI and LiDAR/SAR data, resulting in limited adaptability and performance when facing more complex scenes.

Different from the above methods, our HMDA-DANet model is a cross-domain end-to-end classification model with three advantages. First, it is the first time to develop an HAMAE structure without pre-training for cross-domain classification of MSRS data, providing a solution for some special scenarios where it is difficult to obtain the pre-trained models or there is a lack of a large amount of MSRS data for pre-training. Second, an IAEIEI block is designed to enhance the key intramodality information and explore the intermodality complementarity for more accurate multimodal multiscale feature extraction and cross-modal reconstruction. An MMFF block is further designed to model the long-range spatial and scale dependencies, obtaining the representative multimodal multiscale fusion features. Finally, the DDA block is built to align the semantic and global structural information in each class. Therefore, the powerful multimodal multiscale representation, fine-grained domain adaptation, and lightweight end-to-end classification are the key factors to enable HMDA-DANet to construct the cross-modal, cross-scale, and cross-domain associations, and effectively eliminate the domain gaps between the SD and TD, thus achieving the optimal performance for cross-domain classification of MSRS data.

*1) Results and Analysis on the H13-H18 Dataset:* The quantitative results of all comparison methods and the proposed HMDA-DANet framework are shown in Table I with their classification maps in Fig. 8. Due to the lack of multimodal interaction, SCLUDA obtains the worst classification results, with only 70.14% in OA value; while MLUDA produces 78.29% in OA value, obtaining 8.15% and 0.59% gains over SCLUDA and MSDA, respectively. Among the other five comparison methods, FrF-SSDA introduces a fractional fusion module to overcome the spectral shift of HSIs; however, its preprocessed process will occupy many computing and storage space, and the feature extraction process fails to fully use multimodal complementarity, resulting in relatively low

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TABLE I
CLASSIFICATION ACCURACY (%) USING DIFFERENT UDA METHODS FOR THE H13-H18 DATASET

| Class | SCLUDA | MSDA | MLUDA | FrF-SSDA | MDA-NET | HighDAN | SASS | Proposed |
|---|---|---|---|---|---|---|---|---|
| 1 | 85.89 ± 11.52 | **97.22 ± 1.26** | 68.36 ± 19.55 | 73.60 ± 24.91 | 79.01 ± 11.49 | 71.43 ± 12.12 | 61.31 ± 9.91 | 83.10 ± 4.96 |
| 2 | 56.24 ± 11.74 | 41.67 ± 4.29 | 75.02 ± 14.22 | 57.22 ± 14.25 | 74.85 ± 6.54 | 62.90 ± 6.16 | **86.30 ± 3.87** | 55.67 ± 5.49 |
| 3 | 83.81 ± 6.82 | 85.86 ± 6.77 | 85.32 ± 5.43 | 86.75 ± 6.47 | 86.92 ± 2.65 | **94.96 ± 1.83** | 91.21 ± 4.78 | 94.50 ± 1.66 |
| 4 | 77.27 ± 27.27 | 85.45 ± 19.92 | **100.00 ± 0.00** | 92.73 ± 11.85 | 92.10 ± 11.26 | 76.36 ± 5.93 | 95.45 ± 6.43 | **100.00 ± 0.00** |
| 5 | 88.06 ± 3.01 | **91.47 ± 6.10** | 82.15 ± 8.73 | 77.90 ± 7.98 | 40.03 ± 2.28 | 68.53 ± 5.44 | 81.00 ± 15.24 | 76.03 ± 7.88 |
| 6 | 65.49 ± 2.12 | 78.29 ± 2.55 | 77.25 ± 2.87 | 81.04 ± 3.32 | **92.99 ± 4.88** | 85.15 ± 2.65 | 74.84 ± 2.47 | 87.32 ± 1.62 |
| 7 | 82.02 ± 4.07 | 84.34 ± 3.84 | 82.94 ± 2.39 | 77.09 ± 5.58 | **89.58 ± 5.66** | 75.05 ± 4.05 | 79.30 ± 5.84 | 79.05 ± 6.74 |
| OA | 70.14 ± 1.09 | 77.70 ± 1.30 | 78.29 ± 1.77 | 78.05 ± 0.52 | 78.88 ± 0.81 | 80.17 ± 1.96 | 77.32 ± 1.04 | **82.38 ± 0.33** |
| AA | 76.97 ± 3.22 | 80.61 ± 3.52 | 81.58 ± 3.04 | 78.05 ± 5.68 | 79.36 ± 2.97 | 76.34 ± 2.55 | 81.34 ± 0.81 | **82.24 ± 0.73** |
| $\kappa \times 100$ | 56.95 ± 1.23 | 65.82 ± 1.67 | 66.84 ± 2.49 | 65.03 ± 0.44 | 66.13 ± 2.33 | 68.49 ± 2.61 | 65.82 ± 1.29 | **71.43 ± 0.59** |
| Time (s) | 5.28 | 7.54 | 6.69 | 4.27 | 6.52 | 2.14 | 7.47 | 2.81 |

TABLE II
CLASSIFICATION ACCURACY (%) USING DIFFERENT UDA METHODS FOR THE NA-HA DATASET

| Class | SCLUDA | MSDA | MLUDA | FrF-SSDA | MDA-NET | HighDAN | SASS | Proposed |
|---|---|---|---|---|---|---|---|---|
| 1 | 93.88 ± 2.21 | 94.82 ± 1.36 | 90.68 ± 1.23 | 95.35 ± 0.11 | **99.94 ± 0.07** | 96.77 ± 1.78 | 94.56 ± 1.32 | 94.51 ± 0.62 |
| 2 | 94.98 ± 10.88 | 92.93 ± 5.39 | **100.00 ± 0.00** | 82.93 ± 6.06 | 97.91 ± 1.20 | 99.97 ± 0.02 | **100.00 ± 0.00** | 100.00 ± 0.01 |
| 3 | **100.00 ± 0.00** | **100.00 ± 0.00** | **100.00 ± 0.00** | **100.00 ± 0.00** | 83.59 ± 9.01 | **100.00 ± 0.00** | **100.00 ± 0.00** | **100.00 ± 0.00** |
| 4 | **100.00 ± 0.00** | 99.26 ± 0.25 | 99.63 ± 0.17 | 96.47 ± 0.06 | 93.27 ± 9.33 | 84.84 ± 15.45 | **100.00 ± 0.00** | 94.70 ± 5.09 |
| 5 | 54.39 ± 14.37 | **94.39 ± 1.94** | 91.11 ± 4.44 | 80.85 ± 3.50 | 85.80 ± 7.75 | 76.15 ± 9.59 | 87.24 ± 2.96 | 88.95 ± 4.75 |
| 6 | 91.15 ± 2.24 | 88.03 ± 1.27 | 89.51 ± 1.99 | 90.85 ± 1.97 | 82.70 ± 2.27 | 94.17 ± 7.08 | 97.32 ± 2.06 | **98.72 ± 0.30** |
| 7 | 97.83 ± 1.93 | 23.58 ± 4.28 | 63.00 ± 22.24 | 37.38 ± 30.90 | **100.00 ± 0.00** | 81.49 ± 13.41 | 50.08 ± 27.98 | 75.65 ± 11.13 |
| OA | 90.00 ± 1.60 | 92.91 ± 0.44 | 92.33 ± 0.64 | 91.13 ± 0.52 | 93.25 ± 2.02 | 94.28 ± 0.70 | 94.53 ± 0.63 | **95.33 ± 0.83** |
| AA | 90.32 ± 1.16 | 84.71 ± 0.89 | 90.56 ± 2.91 | 83.40 ± 4.32 | 91.89 ± 2.11 | 90.48 ± 0.54 | 89.89 ± 3.35 | **93.22 ± 1.86** |
| $\kappa \times 100$ | 86.30 ± 2.18 | 90.14 ± 0.62 | 89.50 ± 0.86 | 87.54 ± 0.73 | 90.45 ± 2.85 | 92.00 ± 0.98 | 92.46 ± 0.86 | **93.50 ± 1.14** |
| Time (s) | 3.04 | 5.46 | 4.89 | 4.43 | 17.65 | 4.03 | 6.01 | 4.86 |

TABLE III
CLASSIFICATION ACCURACY (%) USING DIFFERENT UDA METHODS FOR THE H13-TO DATASET

| Class | SCLUDA | MSDA | MLUDA | FrF-SSDA | MDA-NET | HighDAN | SASS | Proposed |
|---|---|---|---|---|---|---|---|---|
| 1 | **99.95 ± 0.09** | 99.84 ± 0.35 | 99.14 ± 0.67 | 99.78 ± 0.36 | 98.41 ± 0.72 | 98.07 ± 1.17 | 99.02 ± 0.61 | 98.45 ± 1.01 |
| 2 | 57.78 ± 10.40 | 50.02 ± 36.36 | 78.18 ± 4.52 | 91.97 ± 3.82 | 72.18 ± 12.84 | 90.32 ± 5.87 | 93.84 ± 2.88 | **93.97 ± 5.66** |
| 3 | 19.54 ± 25.71 | 71.44 ± 40.40 | 39.42 ± 22.72 | 85.72 ± 17.46 | 74.88 ± 17.77 | 40.42 ± 23.49 | 74.49 ± 9.59 | **86.05 ± 5.35** |
| 4 | 84.58 ± 4.23 | 88.17 ± 8.30 | 83.03 ± 10.65 | 80.15 ± 4.62 | 88.94 ± 9.46 | 83.22 ± 8.49 | 84.96 ± 4.70 | **91.41 ± 4.11** |
| OA | 89.30 ± 0.89 | 89.93 ± 3.21 | 92.00 ± 1.68 | 95.13 ± 0.30 | 91.01 ± 3.03 | 93.13 ± 0.47 | 95.39 ± 0.74 | **96.34 ± 0.79** |
| AA | 65.46 ± 5.61 | 77.37 ± 6.44 | 74.94 ± 5.75 | 89.41 ± 3.89 | 83.60 ± 6.72 | 78.01 ± 4.53 | 88.08 ± 2.82 | **92.44 ± 1.15** |
| $\kappa \times 100$ | 78.61 ± 1.75 | 79.89 ± 6.31 | 84.21 ± 3.38 | 90.36 ± 0.60 | 82.12 ± 5.96 | 86.40 ± 0.79 | 90.96 ± 1.46 | **92.84 ± 1.51** |
| Time (s) | 1.94 | 2.79 | 2.42 | 2.30 | 4.96 | 1.11 | 2.83 | 2.79 |

TABLE IV
CLASSIFICATION ACCURACY (%) USING DIFFERENT UDA METHODS FOR THE TO-MU DATASET

| Class | SCLUDA | MSDA | MLUDA | FrF-SSDA | MDA-NET | HighDAN | SASS | Proposed |
|---|---|---|---|---|---|---|---|---|
| 1 | 83.50 ± 3.78 | 86.05 ± 4.06 | 77.61 ± 3.15 | 72.19 ± 7.72 | **98.65 ± 0.77** | 85.89 ± 3.81 | 90.51 ± 9.17 | 88.17 ± 2.79 |
| 2 | 42.88 ± 5.86 | 46.61 ± 10.21 | 50.42 ± 10.06 | 55.56 ± 8.58 | 53.72 ± 18.64 | **91.45 ± 10.04** | 30.36 ± 18.68 | 69.50 ± 5.50 |
| 3 | 57.35 ± 17.02 | 70.84 ± 9.16 | 87.83 ± 6.01 | 87.61 ± 9.92 | 48.37 ± 7.81 | 41.67 ± 6.14 | **91.78 ± 5.66** | 67.56 ± 3.61 |
| OA | 71.63 ± 1.75 | 76.41 ± 3.41 | 74.78 ± 2.95 | 72.18 ± 2.34 | 80.05 ± 0.65 | 78.67 ± 2.07 | 80.19 ± 3.52 | **81.56 ± 2.09** |
| AA | 61.24 ± 5.67 | 67.83 ± 6.16 | 71.95 ± 3.52 | 71.79 ± 2.46 | 64.30 ± 2.72 | 73.00 ± 3.88 | 70.88 ± 3.84 | **75.08 ± 2.65** |
| $\kappa \times 100$ | 46.09 ± 5.42 | 54.21 ± 7.83 | 56.14 ± 4.65 | 52.73 ± 1.76 | 62.38 ± 1.39 | 61.47 ± 3.48 | 63.61 ± 4.61 | **64.23 ± 4.03** |
| Time (s) | 3.36 | 5.11 | 4.49 | 2.80 | 5.82 | 0.96 | 2.79 | 2.24 |

accuracy (i.e., 78.05% in OA and AA values). MDA-NET and SASS improve the cross-domain classification accuracy overall in OA, AA, and $\kappa$ values than FrF-SSDA. As a semantic segmentation algorithm, HighDAN ranks second with OA value 80.17%; however, the lack of specialized cross-modal design limits its performance in other metrics (i.e., AA and $\kappa$).

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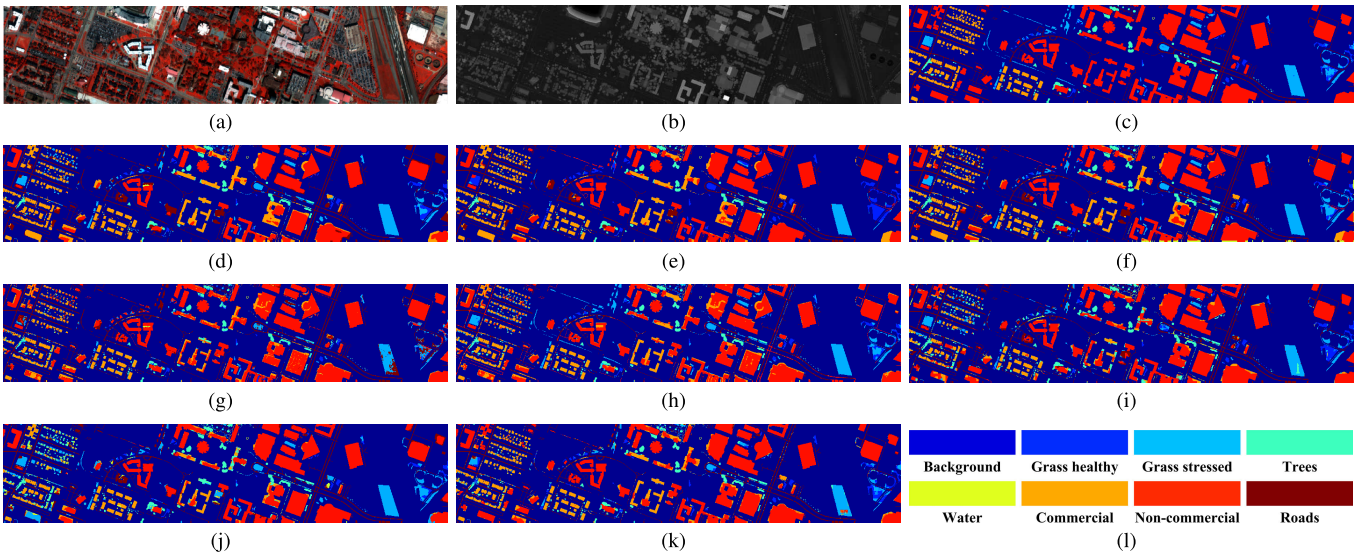



Fig. 8. Classification maps for the H13-H18 dataset. (a) Target HSI (Rband: 35, Gband: 15, Bband: 10). (b) Target LiDAR. (c) Target ground-truth map. (d) SCLUDA. (e) MSDA. (f) MLUDA. (g) FrF-SSDA. (h) MDANet. (i) HighDAN. (j) SASS. (k) Proposed. (l) Legend.

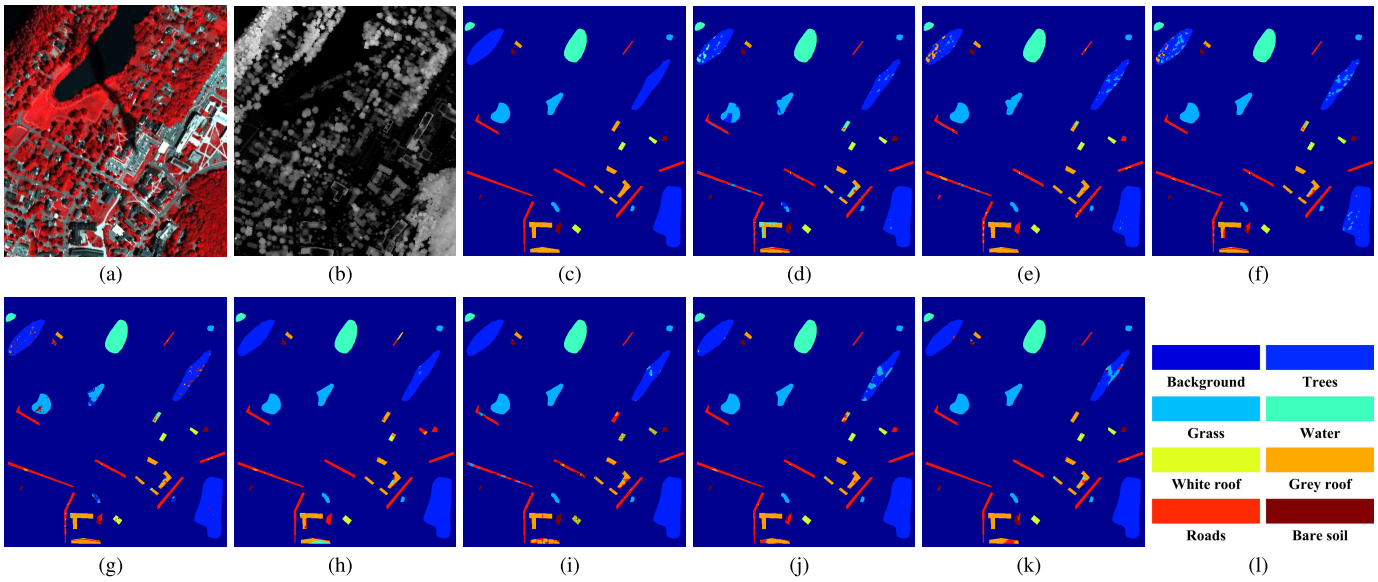



Fig. 9. Classification maps for the NA-HA dataset. (a) Target HSI (Rband: 20, Gband: 35, Bband: 15). (b) Target LiDAR. (c) Target ground-truth map. (d) SCLUDA. (e) MSDA. (f) MLUDA. (g) FrF-SSDA. (h) MDANet. (i) HighDAN. (j) SASS. (k) Proposed. (l) Legend.

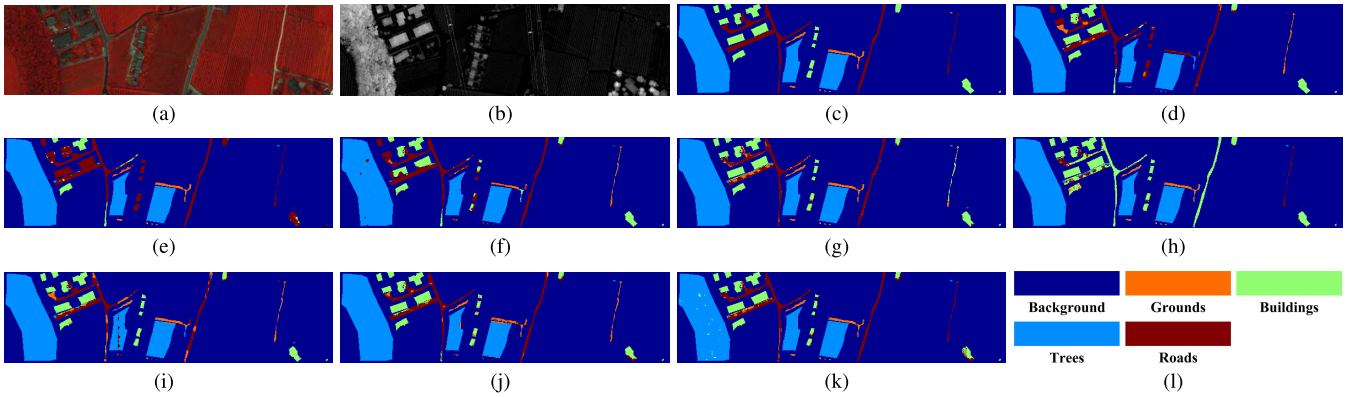



Fig. 10. Classification maps for the H13-TO dataset. (a) Target HSI (Rband: 42, Gband: 20, Bband: 5). (b) Target LiDAR. (c) Target ground-truth map. (d) SCLUDA. (e) MSDA. (f) MLUDA. (g) FrF-SSDA. (h) MDANet. (i) HighDAN. (j) SASS. (k) Proposed. (l) Legend.

Differently, HMDA-DANet generates at least 2.21% gain in OA, 0.90% in AA, and 2.94% in $\kappa$ values, all of which reach the optimal level. In addition, from Fig. 8, it generates the visually better results than other methods. Experimental

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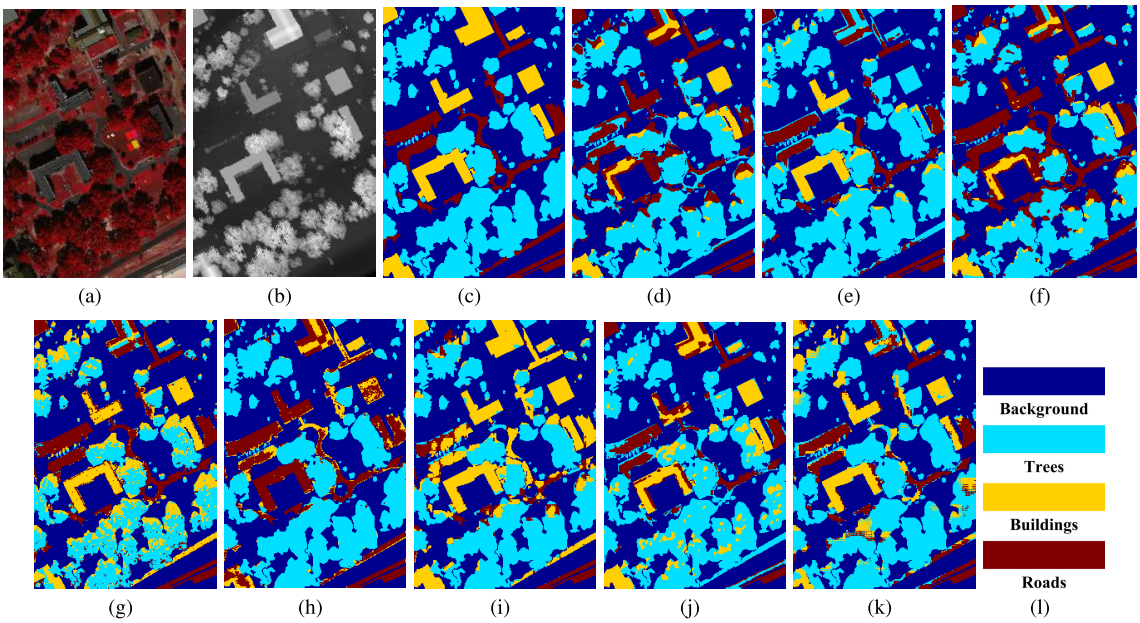



Fig. 11. Classification maps for the TO-MU dataset. (a) Target HSI (Rband: 45, Gband: 25, Bband: 15). (b) Target LiDAR. (c) Target ground-truth map. (d) SCLUDA. (e) MSDA. (f) MLUDA. (g) FrF-SSDA. (h) MDANet. (i) HighDAN. (j) SASS. (k) Proposed. (l) Legend.

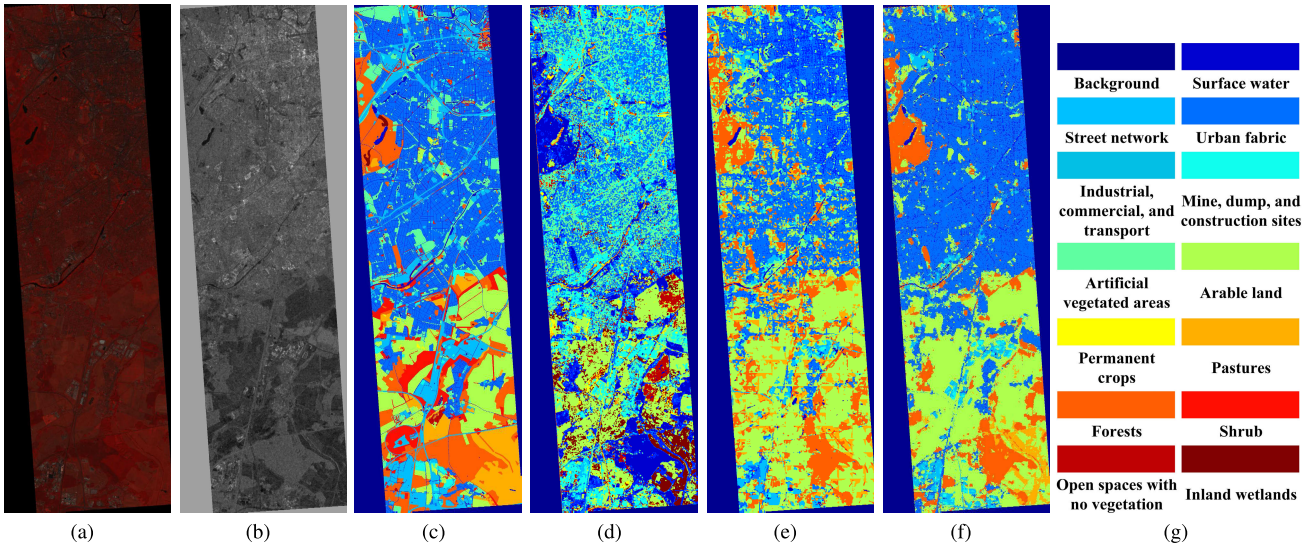



Fig. 12. Classification maps for the C2Seg-AB dataset. (a) Target HSI (Rband: 70, Gband: 30, Bband: 10). (b) Target SAR. (c) Target ground-truth map. (d) SASS. (e) HighDAN. (f) Proposed. (g) legend.

results verify the advantages of the proposed HMDA-DANet framework in cross-domain classification of MSRS data.

*2) Results and Analysis on the NA-HA Dataset:* Table II and Fig. 9 report the classification and visualization results of different methods for the NA-HA dataset, respectively. From Table II, the overall classification performance of SCLUDA and FrF-SSDA is the worst, with OA value only 90.00% and 91.13%, respectively. MSDA, MLUDA, and MDA-NET obtain the third best classification performance in OA value; however, the specialized multimodal structure design enables MDA-NET to obtain better performance in terms of AA and $\kappa$ values. Furthermore, HighDAN and SASS improve the OA value to over 94.28% due to their better learning strategy. As expected, our HMDA-DANet achieves the highest performance in all three metrics, where the classification accuracy exceeds 75.65% across all classes, making the AA value the highest among all methods. In the other two metrics, its accuracy is at least 0.80% higher than the highest values obtained by SASS. Moreover, as shown in Fig. 9, its classification map exhibits less mislabeled areas and clearer boundary delineation between different classes, which is the closest to the ground-truth map. These experimental results further prove that our HMDA-DANet performs better than other comparison methods for cross-domain classification of NSRS data.

*3) Results and Analysis on the H13-TO Dataset:* Table III illustrates the classification accuracy of all methods for the H13-TO dataset. Specifically, the OA values of SCLUDA and MSDA are both below 90.00%, ranking the worst among all methods. MLUDA, MDA-NET, and HighDAN increase the OA value by at least 1.08%; however, the limited DA

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TABLE V
CLASSIFICATION ACCURACY (%) ON THE C2SEG-AB DATASET

| Class | SASS | HighDAN | Proposed |
|---|---|---|---|
| 1 | 10.03 ± 4.38 | **14.62 ± 14.88** | 13.91 ± 9.88 |
| 2 | 6.58 ± 8.10 | 6.01 ± 6.94 | **9.09 ± 6.08** |
| 3 | 39.75 ± 18.61 | 77.48 ± 17.27 | **84.53 ± 4.27** |
| 4 | 12.36 ± 6.86 | 17.79 ± 12.75 | **42.39 ± 12.41** |
| 5 | **17.58 ± 7.49** | 0.83 ± 1.85 | 0.97 ± 2.13 |
| 6 | 10.50 ± 12.01 | 14.69 ± 10.23 | **21.29 ± 14.28** |
| 7 | 49.35 ± 28.68 | **89.23 ± 4.77** | 80.09 ± 10.43 |
| 8 | **1.77 ± 3.75** | — | — |
| 9 | 0.19 ± 0.17 | 13.59 ± 20.19 | **36.57 ± 19.80** |
| 10 | 0.75 ± 0.80 | 47.81 ± 17.19 | **66.05 ± 6.53** |
| 11 | **2.49 ± 3.61** | 0.21 ± 0.46 | 0.04 ± 0.05 |
| 12 | **1.12 ± 1.92** | — | — |
| 13 | **2.12 ± 1.99** | — | — |
| OA | 22.23 ± 3.49 | 46.73 ± 1.73 | **55.29 ± 1.72** |
| AA | 11.89 ± 1.48 | 21.71 ± 3.17 | **27.30 ± 0.59** |
| $\kappa \times 100$ | 11.47 ± 2.44 | 33.14 ± 1.66 | **43.77 ± 1.99** |
| Time (s) | 436.60 | 11.12 | 23.81 |

capability prevents them from achieving good performance in AA value. Furthermore, due to better feature extraction and alignment strategies, the OA and AA values are improved to at least 95.13% and 88.08% by FrF-SSDA and SASS, respectively. Especially, our HMDA-DANet improves the classification accuracy of each class to over 86.05% to achieve the best results in almost all classes, thus resulting in an OA value of 96.34%, ranking first. Similar conclusions can also be drawn from the classification maps in Fig. 10, verifying the effectiveness of the proposed HMDA-DANet framework.

*4) Results and Analysis on the TO-MU Dataset:* Although there are only three classes in the TO-MU dataset, it is relatively difficult to interpret it due to different sensor characteristics and complex scene differences. According to Table IV, the OA values of most comparison methods (i.e., SCLUDA, MSDA, MLUDA, FrF-SSDA, and HighDAN) are below 80.00%. With better multimodal learning, MDA-NET and SASS increase the OA value to 80.05% and 80.19%, respectively, achieving at least 1.38% gains than the other five comparison methods. However, most comparison methods generate lower AA value, indicating that their generalization and scene adaptability need to be further improved. Although the classification accuracy of our HMDA-DANet is not optimal for each class, it shows certain advantages in overall performance, with at least 1.37% improvement in OA value compared to the other methods. In addition, the performance on the classification maps shown in Fig. 11 is consistent with that in Table IV, which also proves the superiority of our HMDA-DANet framework.

*5) Results and Analysis on the C2Seg-AB Dataset:* The C2Seg-AB dataset was publicly released in 2023 IEEE WHISPERS C2Seg Challenge [24], [38]. Due to its large size, taking into account the performance of all comparison methods and the limitations of computer memory resources, Table V presents the quantitative results of the proposed HMDA-DANet framework and two representative comparison methods, where principal component analysis is applied to reduce the spectral number for SASS. SASS only generates 22.23% in OA value, 11.89% in AA value, and 11.47% in $\kappa$ value, which may be caused by spectral dimensionality reduction and insufficient training. However, it can identify some special land covers with extremely imbalanced samples, such as Class 8, Class 12, and Class 13. HighDAN achieves the second best classification accuracy, and improves the OA value to 46.73%. In particular, the OA value of our HMDA-DANet reaches 55.29%, which is even comparable to that of HighDAN using three components as inputs [24]. However, the accuracy of identifying the above special classes is still limited, and there are also cases where some classes fail to be identified at all. From Fig. 12, the classification map of HMDA-DANet is smoother. Experimental results to some extent verify that the end-to-end classification may be an effective way to better explore the intrinsic information of large-scale MSRS datasets, achieving more satisfactory efficiency and performance.

*6) Complexity Analysis:* To further highlight the efficiency advantages of the proposed HMDA-DANet framework, the last lines of Tables I–V also report and compare the average testing time (s) of all methods. More specifically, compared to the patch-based methods (i.e., SCLUDA, MSDA, MLUDA, FrF-SSDA, MDA-NET, SASS), our HMDA-DANet shows better advantages in four scenes (i.e., H13-H18, NA-HA, H13-TO, and TO-MU datasets), with absolute performance and efficiency advantages in larger scale scene (i.e., C2Seg-AB dataset). Moreover, compared with HighDAN, which is also a cross-domain end-to-end classification method, the use of full-band spectral information and the strategy of cross-modal reconstruction limit the efficiency of HMDA-DANet to a certain extent. However, by comprehensively analyzing the classification performance, inference efficiency, and scene generalization, our HMDA-DANet framework clearly performs more satisfactorily, further illustrating its effectiveness and advantages for cross-domain MSRS data classification.

### E. Ablation Study

To measure the contributions of the FDLConv, IAEIEI, MMFF, and DDA blocks to cross-domain classification performance of our HMDA-DANet, the third experiment is conducted for a detailed ablation study. In particular, the Baseline model is considered, where only the HAMAE structure with the direct concatenation operation that is used to replace the MMFF block is executed for multimodal multiscale feature extraction and fusion, whose results are fed into the end-to-end classification head. The proposed HMDA-DANet with and without each component is denoted as proposed (with) and proposed (without), respectively. Table VI reports the experimental results (i.e., OA and $\kappa$) for the five cross-domain MSRS datasets. For the sake of comparative fairness, no pretrained models or processes are provided for all experiments.

*1) Analysis of the EC Discriminator:* The EC discriminator is designed to align the global structural information in the prediction space between the SD and TD, whose contributions on our HMDA-DANet are analyzed in the first line of parts (b) and (c) of Table VI. On the one hand, the EC discriminator improves the OA value of the Baseline from

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TABLE VI
ABLATION STUDY OF OUR HMDA-DANET ON THE FIVE CROSS-DOMAIN MSRS DATASETS

| | Module | | | | | | H13-H18 Dataset | | NA-HA Dataset | | H13-TO Dataset | | TO-MU Dataset | | C2Seg-AB Dataset | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | DLConv | FDLConv | IAEIEI | MMFF | FG | EC | OA | $\kappa \times 100$ | OA | $\kappa \times 100$ | OA | $\kappa \times 100$ | OA | $\kappa \times 100$ | OA | $\kappa \times 100$ |
| (a) | ✘ | ✘ | ✘ | ✘ | ✘ | ✘ | 7.79 | — | 14.88 | — | 56.22 | — | 33.14 | — | 8.02 | — |
| (b) | ✘ | ✘ | ✘ | ✘ | ✘ | ✔ | 77.95 | 65.56 | 93.11 | 90.37 | 82.09 | 68.82 | 71.11 | 49.33 | 33.35 | 22.24 |
| | ✘ | ✘ | ✘ | ✘ | ✔ | ✘ | 75.51 | 62.05 | 93.42 | 90.92 | 84.02 | 71.90 | 72.74 | 48.82 | 32.06 | 21.30 |
| | ✘ | ✘ | ✘ | ✘ | ✔ | ✔ | 79.15 | 66.46 | 93.75 | 91.35 | 91.65 | 83.74 | 73.28 | 49.43 | 41.73 | 22.23 |
| (c) | ✘ | ✔ | ✔ | ✔ | ✔ | ✘ | 80.63 | 69.65 | 94.38 | 92.15 | 93.10 | 86.81 | 77.25 | 59.69 | 47.59 | 35.93 |
| | ✘ | ✔ | ✔ | ✔ | ✘ | ✔ | 80.49 | 68.15 | 94.46 | 92.25 | 93.38 | 86.94 | 76.11 | 56.75 | 45.96 | 31.96 |
| | ✘ | ✔ | ✔ | ✔ | ✘ | ✘ | 15.32 | — | 15.78 | — | 59.15 | — | 37.64 | — | 9.15 | — |
| | ✘ | ✔ | ✔ | ✘ | ✔ | ✔ | 80.60 | 69.31 | 94.71 | 92.67 | 94.21 | 88.76 | 79.72 | 61.87 | 52.52 | 38.56 |
| | ✘ | ✔ | ✘ | ✔ | ✔ | ✔ | 80.55 | 69.47 | 94.35 | 92.14 | 93.64 | 87.11 | 78.79 | 61.70 | 52.97 | 40.41 |
| | ✔ | ✘ | ✔ | ✔ | ✔ | ✔ | 81.12 | 68.75 | 94.47 | 92.29 | 94.88 | 89.93 | 80.02 | 63.09 | 52.38 | 40.99 |
| | ✘ | ✘ | ✔ | ✔ | ✔ | ✔ | 81.42 | 69.95 | 94.18 | 91.90 | 95.16 | 90.52 | 79.84 | 63.05 | 52.56 | 39.06 |
| (d) | ✘ | ✔ | ✔ | ✔ | ✔ | ✔ | **82.38** | **71.43** | **95.33** | **93.50** | **96.34** | **92.84** | **81.56** | **64.23** | **55.29** | **43.77** |

7.79%, 14.88%, 56.22%, 33.14%, and 8.02% to 77.95%, 93.11%, 82.09%, 71.11%, and 33.35% for the five datasets, respectively. On the other hand, compared with the proposed (without EC), HMDA-DANet obtains 1.75%, 0.95%, 3.24%, 4.32%, and 7.70% gain in OA value, respectively, verifying that the global structural information alignment of the EC discriminator is beneficial for cross-domain classification of MSRS data.

*2) Analysis of the FG Discriminator:* The FG discriminator is further built to overcome the domain gaps by aligning the multimodal features in each class. We compare the above design choice with two variants, i.e., Baseline+FG and the proposed (without FG), as shown in the second line of parts (b) and (c) of Table VI. We can draw conclusions similar to those of the EC discriminator, verifying the effectiveness of class-related semantic feature alignment from the FG discriminator for cross-domain MSRS data classification.

*3) Analysis of the DDA Block:* By integrating the advantages of the EC and FG discriminators, a DDA block is developed for cross-domain alignment. We run the Baseline and HMDA-DANet models with or without the DDA block to analyze its superiority, as shown in the third line of parts (b) and (c) of Table VI. Compared with the Baseline, the DDA block noticeably improves the OA value from 7.79%, 14.88%, 56.22%, 33.14%, and 8.02% to 79.15%, 93.75%, 91.65%, 73.28%, and 41.73% for the five datasets; while from 15.32%, 15.78%, 59.15%, 37.64%, and 9.15% to 82.38%, 95.33%, 96.34%, 81.56%, and 55.29% for HMDA-DANet, respectively. Moreover, compared with the EC or FG discriminator, the DDA block improves the OA value by at least 0.33% for Baseline, and 0.87% for HMDA-DANet in the five datasets.

In summary, although the EC and FG discriminators perform differently in the five MSRS scenes, they show varying degrees of positive impacts on the cross-domain classification performance of Baseline and HMDA-DANet models. They do not rely on specific datasets or models. Moreover, by integrating the advantages of these two discriminators, the DDA block can not only accurately distinguish domains from the perspective of class-level semantic feature alignment, but also learn the potential structural relationships of different classes between the SD and TD from the perspective of prediction structural alignment. Experimental results prove the respective advantages of EC and FG discriminators, as well as the effectiveness of the DDA block in eliminating complex domain discrepancy and improving the classification performance of HMDA-DANet for cross-domain MSRS data.

*4) Analysis of the MMFF Block:* The MMFF block is built to model the long-range spatial and scale dependencies between multimodal multiscale features. Experiments are carried out by comparing the proposed HMDA-DANet framework with its variant, i.e., the proposed (without MMFF). From the fourth line of part (c) in Table VI, the MMFF block provides at least 0.62% gains in OA value, which verifies its effectiveness in long-range correlation modeling and fusion of multimodal multiscale features for cross-domain joint classification.

*5) Analysis of the IAEIEI Block:* It is built for cross-modal reconstruction by exploring the complementary information between multimodal features using unsupervised training. We run HMDA-DANet with or without the IAEIEI block to study its impacts, whose results are given in the fifth line of part (c) of Table VI. When the IAEIEI block is introduced, the OA value of HMDA-DANet is improved by 1.83%, 0.98%, 2.70%, 2.77%, and 2.32% for the five datasets, respectively, proving the effectiveness of the IAEIEI block in cross-modal reconstruction and cross-domain classification of MSRS data.

*6) Analysis of the FDLConv Block:* A new frequency attention (i.e., FDLConv block) is designed as a frequency domain filter in our HMDA-DANet. By dynamically modulating frequency components, some unimportant frequency components are removed, making the whole framework focus more on some important structural information of MSRS data. We conduct the experiments to analyze the influence of FDLConv block on the cross-domain classification performance of HMDA-DANet, whose results are reported in the last line of part (c) in Table VI. Compared the variant [i.e., the proposed (without FDLConv)], introducing the FDLConv block can improve the OA value of HMDA-DANet by 0.96%, 1.15%, 1.18%, 1.72%, and 2.73% for the five datasets, respectively.

Furthermore, we utilize the DLConv block in the spatial domain to further prove the advantages of frequency attention. By comparing with the results in the last three lines of

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Table VI, introducing frequency attention can, to some extent, enable the whole model to focus on some important structural information in MSRS data and avoid some interference information that affect cross-modal reconstruction, thus improving the feature extraction ability and classification performance of the proposed HMDA-DANet framework. Experimental results verify the effectiveness of the designed FDLConv block.

## V. Conclusion

In this article, a UDA framework (i.e., HMDA-DANet) has been proposed for cross-domain end-to-end classification of MSRS data. An HAMAE structure is first developed, where FDLConv and IAEIEI blocks are designed to assist in spatial information enhancement, multimodal multiscale feature extraction, and cross-modal construction from MSRS data. Then, an MMFF block is designed to model the long-range spatial and scale dependencies of these features. A DDA block is further constructed, in conjunction with HAMAE, to model the cross-modal, cross-scale, and cross-domain associations, thus generating the domain-invariant multimodal fusion features for cross-scene end-to-end classification. Plenty of comparison and ablation experiments have been conducted on the five cross-domain MSRS datasets, whose results illustrate that the proposed HMDA-DANet can consistently provide better classification performance over other state-of-the-art methods.

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